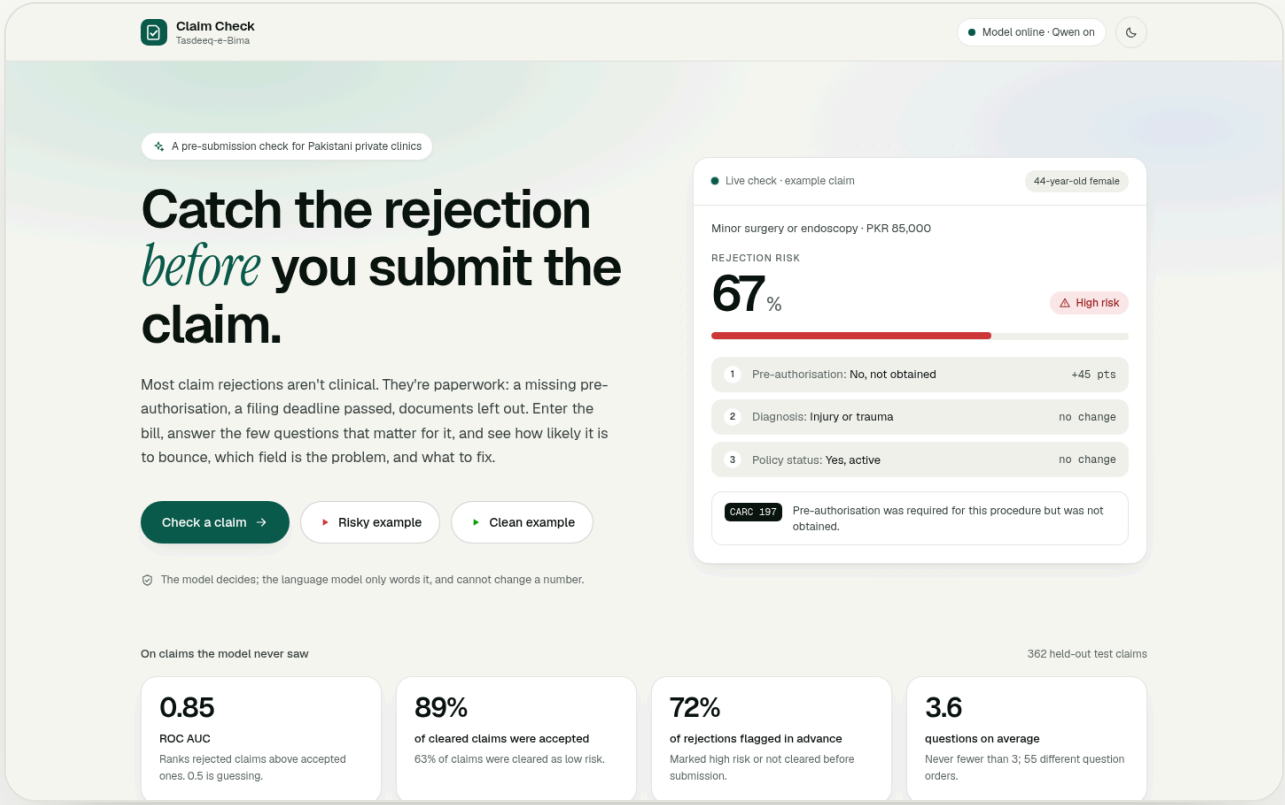
BANO QABIL AI HACKATHON

# Claim Check

Catch the insurance-claim rejection *before* the claim is sent.





Mirza Ahsan Baig · Safee Akmal

Tasdeeq-e-Bima

THE PROBLEM

# Claims get rejected for paperwork, not medicine

- Weeks after treatment the claim bounces: **no pre-authorisation**, **missing documents**, a **missed deadline**
- The clinic reworks it and waits again — small clinics stop taking insured patients

**15%**

of US private-insurer claims denied first time[1]

**$25.7B**

spent by US providers fighting denials (2023)[1]

**86%**

of denials potentially avoidable[2]

**71%**

of India's health-claim value actually paid (FY24)[3]

Sources: [1] Premier Inc. · [2] Becker's Hospital Review · [3] Outlook Business

CASE STUDY · JUBILEE HEALTH INSURANCE, KENYA

# One insurer, every way a claim can fail

### THE PATH EVERY CLAIM MUST SURVIVE

1. **Verify** the medical card and national ID
2. **Pre-authorise** 48 h before a planned admission[4]
3. **Claim form** signed by patient and doctor
4. **Submit** every bill, report and summary in time
5. **Vetting**, then payment — if nothing is missing

| JUBILEE CAN REJECT FOR… | CLAIM CHECK, BEFORE SENDING |
|---|---|
| No pre-authorisation | **Asked first, every time** |
| Incomplete documents | **Flags missing papers** |
| Discharge summary ≠ approval | **Cross-checks the papers** |
| Clinic not on the plan's panel | **Asks panel status** |
| Service not covered by the plan | **Plan-coverage check** |
| Filed too late | **Filing-deadline check** |
| *Pre-existing condition* | *Not yet — roadmap* |

The cost of getting it wrong: Nairobi Hospital suspended Jubilee in 2020 over a claimed **KSh 357 million** in unpaid claims dating back to 2018.[5]

Sources: [4] Jubilee Insurance Kenya · [5] Business Daily Africa

OUR SOLUTION

# A spell-checker for insurance claims

1 **Photograph the papers**

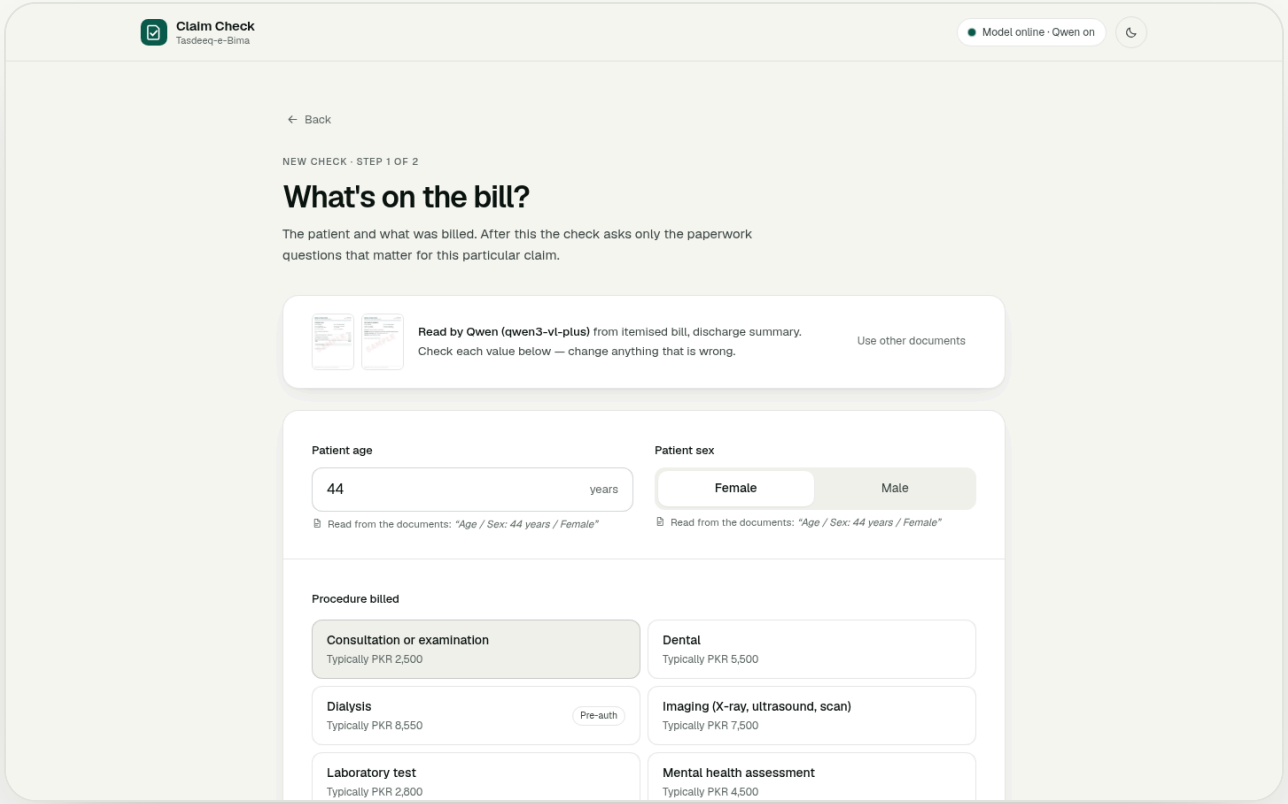
Qwen reads them; the clerk confirms

2 **Answer what they can’t**

Three or four questions, chosen for this claim

3 **Fix before sending**

Chance of rejection, the cause, the fix



HOW IT WORKS

# Qwen reads and explains. The model decides.

**QWEN-VL · READS**

Fills in what is printed; the clerk confirms every value

**MODEL + RULES · DECIDE**

Risk, verdict, reason code, insurer and document checks

**QWEN · EXPLAINS**

One plain sentence, checked before it is shown

- Asks only what the papers can’t settle — pre-authorisation is **always** asked
- **3.6** questions on average, in **55** different orders
- Ends with the fix, the official rejection code and a one-page printout

RESULTS

# Tested on claims it never saw

**89%**

of claims it cleared were accepted

**72%**

of rejections flagged before submission

**56%**

of “high risk” claims were rejected (average 24%)

**0.85**

AUC with every answer known (0.76 mid-check)

362 held-out simulated claims, used once. Verdict thresholds were fitted on a separate split, not chosen by hand.

BUSINESS CASE

# Everyone in the claim wins

### CLINICS

- Paid sooner
- Less rework
- Predictable cash flow

### INSURERS & TPAS

- Cleaner claims
- Fewer appeals
- Lower admin cost

### PATIENTS

- Clinics keep accepting their insurance
- Care isn’t held up by paperwork

**How it earns (proposed):** licence to TPAs for their whole panel · clinic subscription · national e-claims schemes

HONEST ABOUT IT

# Limits today, and what comes next

### WHAT IT DOES NOT DO

- Train on real claims — none are public; the rules and codes are real
- Replace staff — it is a second check
- Submit claims or learn by itself

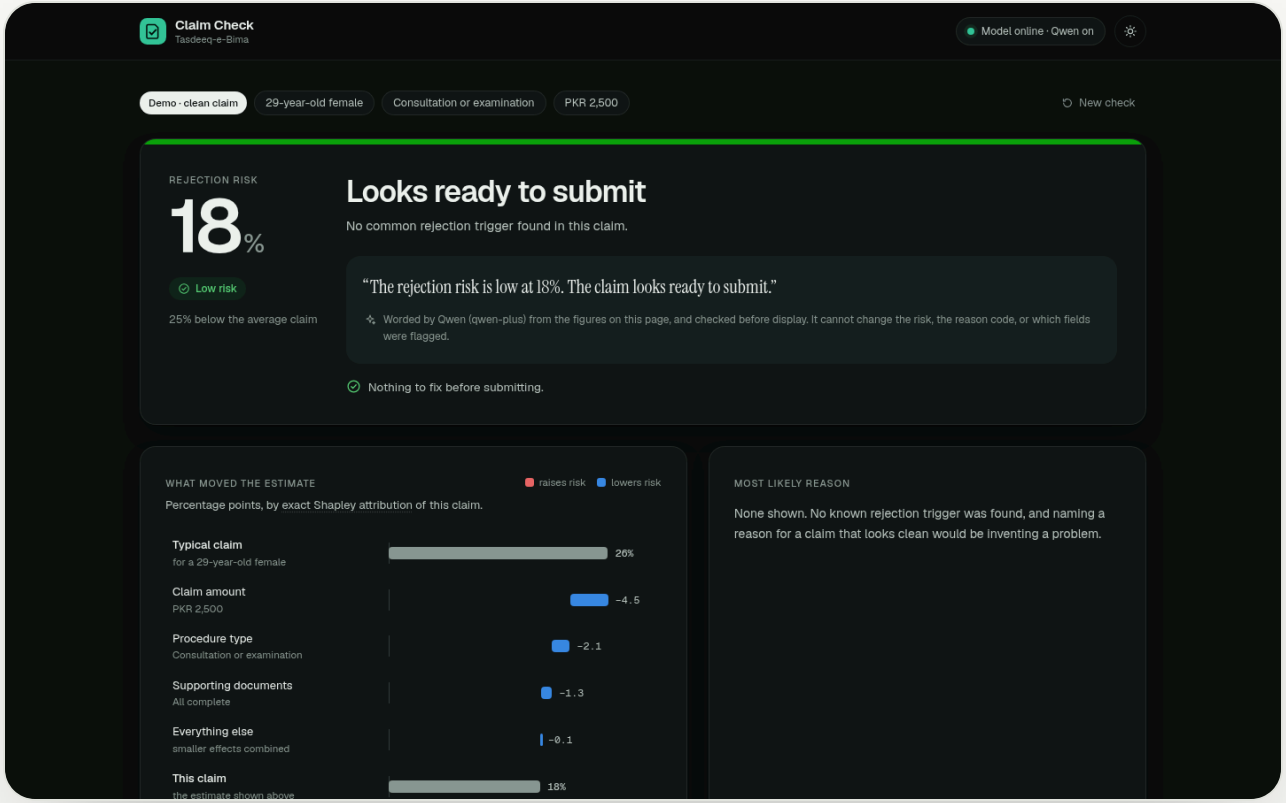
### WHAT WE'D BUILD NEXT

- Pilot with a clinic or TPA on real claims
- Per-insurer rules — Jubilee's first
- Login, encryption, links to billing systems

CLAIM CHECK · TASDEEQ-E-BIMA

# Catch the rejection before you submit.

We're looking for a **partner clinic or TPA** to pilot with real claims.




Mirza Ahsan Baig · Safee Akmal

Thank you

SOURCES

# Where the figures come from


1. Premier Inc., “Claims Adjudication Costs Providers $25.7 Billion”, Feb 2025 (2023 data)
   premierinc.com/newsroom/blog/claims-adjudication-costs-providers-25-7-billion
2. Becker’s Hospital Review, on the Change Healthcare 2020 Revenue Cycle Denials Index (102M claims, 1,500 hospitals)
   beckershospitalreview.com/finance/86-of-denials-are-potentially-avoidable-strategies-to-better-prevent-manage-denials
3. Outlook Business, on the IRDAI Annual Report 2023-24, 30 Dec 2024
   outlookbusiness.com/economy-and-policy/health-insurers-disallowed-claims-worth-rs-15100-crore-during-fy24
4. Jubilee Insurance Kenya, CoverBora brochure (pre-authorisation terms)
   jubileeinsurance.com/ke/wp-content/uploads/2024/01/CoverBora-Brochure.pdf
5. Business Daily Africa, “Jubilee Insurance, Nairobi Hospital in debt row”, June 2020
   businessdailyafrica.com/bd/news/jubilee-insurance-nairobi-hospital-in-debt-row-2293534


Our own results: models/metadata.json and docs/model-card.md in the project repository (3,612 simulated claims, test split of 362).